

Exercise - Create resources in Azure Cosmos DB for NoSQL using .NET :

```
portazurc@#home
Microsoft Azure Upgrade
Search resources, services, and docs (G+/) Copilot
mayureshkpatil50@gmail...
DEFAULT DIRECTORY (MAYURES...)

bash
mayuresh [ ~ ]$ az accountName=cosmosexercise$RANDOM
mayuresh [ ~ ]$ az cosmosdb create \
--name $accountName \
--resource-group $resourceGroup \
--kind GlobalDocumentDB \
--locations regionName=westus
enable_gbs is not a known attribute of class <class 'azure.mgmt.cosmosdb.models._models_py3.DatabaseAccountCreateUpdateParameters'> and will be ignored
{
  "analyticalStorageConfiguration": {
    "schemaType": "WellDefined"
  },
  "apiProperties": null,
  "backupPolicy": {
    "migrationState": null,
    "periodicModeProperties": {
      "backupIntervalInMinutes": 240,
      "backupRetentionIntervalInHours": 8,
      "backupStorageRedundancy": "Geo"
    },
    "type": "Periodic"
  },
  "capabilities": [],
  "capacity": null,
  "connectorOffer": null,
  "consistencyPolicy": {
    "defaultConsistencyLevel": "Session",
    "maxIntervalInSeconds": 5,
    "maxStalenessPrefix": 100
  },
  "cors": [],
  "createMode": null,
  "customerManagedKeyStatus": null,
  "databaseAccountOfferType": "Standard",
  "defaultIdentity": "FirstPartyIdentity",
  "defaultPriorityLevel": "High",
  "disableKeyBasedMetadataWriteAccess": false,
  "disableLocalAuth": false,
  "documentEndpoint": "https://cosmosexercise13342.documents.azure.com:443/",
  "enableAnalyticalStorage": false,
  "enableAutomaticFailover": true,
  "enableBurstCapacity": false,
  "enableCassandraConnector": null,
  "enableFreeTier": false,
  "enableMultipleWriteLocations": false,
  "enablePartitionMerge": false,
  "enablePerRegionPerPartitionAutoscale": true,
  "enablePriorityBasedExecution": false,
```

```
< > portal.azure.com/#home
Microsoft Azure Upgrade Search resources, services, and docs (G+) Copilot
Bash ? {}
},
"tags": {},
"type": "Microsoft.DocumentDB/databaseAccounts",
"virtualNetworkRules": [],
"writeLocations": [
  {
    "documentEndpoint": "https://cosmosexercice13342-westus.documents.azure.com:443/",
    "failoverPriority": 0,
    "id": "Cosmosexercice13342-westus",
    "isZoneRedundant": false,
    "locationName": "West US",
    "provisioningState": "Succeeded"
  }
]
}
mayuresh [ ~ ]$ az cosmosdb show --name $accountName \
--resource-group $resourceGroup \
--query 'documentEndpoint' --output tsv
https://cosmosexercice13342.documents.azure.com:443/
mayuresh [ ~ ]$ az cosmosdb keys list --name $accountName \
--resource-group $resourceGroup \
--query 'primaryKey' --output tsv
[Warning] This output may compromise security by showing secrets. Learn more at: https://go.microsoft.com/fwlink/?linkid=2258669
ry8tC2gVwVMlCD12y1Ez53yJ8kWhhhNEfqTNUV8ogL7qWZU3TKIDw7yXIJvqxYgJHGjEO3woqhCbIT9auu==
mayuresh [ ~ ]$ mkdir cosmosdb
cd cosmosdb
mayuresh [ ~ /cosmosdb ]$ dotnet new console

Welcome to .NET 9.0!
-----
SDK Version: 9.0.308

Telemetry

The .NET tools collect usage data in order to help us improve your experience. It is collected by Microsoft and shared with the community. You can opt-out of telemetry by setting the DOTNET_CLI_TELEMETRY_OPTOUT environment variable to '1' or 'true' using your favorite shell.

Read more about .NET CLI Tools telemetry: https://aka.ms/dotnet-cli-telemetry

-----
Installed an ASP.NET Core HTTPS development certificate.
To trust the certificate, run 'dotnet dev-certs https --trust'
Learn about HTTPS: https://aka.ms/dotnet-https

-----
Write your first app: https://aka.ms/dotnet-hello-world
Find out what's new: https://aka.ms/dotnet-whats-new
```

```
portal.azure.com/#home
Microsoft Azure Upgrade
Search resources, services, and docs (G+/)
Copilot
mayureshkpatil50@gmail.com
DEFAULT DIRECTORY (MAYURESHK)

Processing post-creation actions...
Restoring /home/mayuresh/comosdb/comosdb.csproj:
Restore succeeded.

mayuresh [ ~/comosdb ]$ dotnet add package Microsoft.Azure.Cosmos --version 3.*
dotnet add package Newtonsoft.Json --version 13.*
dotnet add package dotenv.net

Build succeeded in 0.7s
info : X.509 certificate chain validation will use the fallback certificate bundle at '/usr/share/dotnet/sdk/9.0.308/trustedroots/codesignctl.pem'.
info : X.509 certificate chain validation will use the fallback certificate bundle at '/usr/share/dotnet/sdk/9.0.308/trustedroots/timestampctl.pem'.
info : Adding PackageReference for package 'Microsoft.Azure.Cosmos' into project '/home/mayuresh/comosdb/comosdb.csproj'.
info : Restoring packages for /home/mayuresh/comosdb/comosdb.csproj...
info : GET https://api.nuget.org/v3-flatcontainer/microsoft.azure.cosmos/index.json
info : OK https://api.nuget.org/v3-flatcontainer/microsoft.azure.cosmos/index.json 255ms
info : GET https://api.nuget.org/v3-flatcontainer/microsoft.azure.cosmos/3.57.0/microsoft.azure.cosmos.3.57.0.nupkg
info : OK https://api.nuget.org/v3-flatcontainer/microsoft.azure.cosmos/3.57.0/microsoft.azure.cosmos.3.57.0.nupkg 15ms
info : GET https://api.nuget.org/v3-flatcontainer/azure.core/index.json
info : GET https://api.nuget.org/v3-flatcontainer/microsoft.bcl.asyncinterfaces/index.json
info : GET https://api.nuget.org/v3-flatcontainer/system.valuetuple/index.json
info : GET https://api.nuget.org/v3-flatcontainer/system.threading.tasks.extensions/index.json
info : GET https://api.nuget.org/v3-flatcontainer/microsoft.bcl.asyncinterfaces/6.0.0/microsoft.bcl.asyncinterfaces.6.0.0.nupkg
info : GET https://api.nuget.org/v3-flatcontainer/system.unsafe/index.json
info : GET https://api.nuget.org/v3-flatcontainer/system.buffers/index.json
info : GET https://api.nuget.org/v3-flatcontainer/system.numerics.vectors/index.json
info : GET https://api.nuget.org/v3-flatcontainer/system.memory/index.json
info : GET https://api.nuget.org/v3-flatcontainer/system.diagnostics.diagnosticsource/index.json
info : GET https://api.nuget.org/v3-flatcontainer/system.configuration.configurationmanager/index.json
info : GET https://api.nuget.org/v3-flatcontainer/system.collections.immutable/index.json
info : OK https://api.nuget.org/v3-flatcontainer/azure.core/index.json 252ms
info : GET https://api.nuget.org/v3-flatcontainer/azure.core/1.44.1/azure.core.1.44.1.nupkg
info : OK https://api.nuget.org/v3-flatcontainer/microsoft.bcl.asyncinterfaces/6.0.0/microsoft.bcl.asyncinterfaces.6.0.0.nupkg 287ms
info : GET https://api.nuget.org/v3-flatcontainer/microsoft.bcl.asyncinterfaces/6.0.0/microsoft.bcl.asyncinterfaces.6.0.0.nupkg 38ms
info : OK https://api.nuget.org/v3-flatcontainer/microsoft.bcl.asyncinterfaces/6.0.0/microsoft.bcl.asyncinterfaces.6.0.0.nupkg 22ms
info : GET https://api.nuget.org/v3-flatcontainer/system.valuetuple/4.5.0/system.valuetuple.4.5.0.nupkg
info : GET https://api.nuget.org/v3-flatcontainer/system.clientmodel/index.json
info : GET https://api.nuget.org/v3-flatcontainer/system.text.json/index.json
info : GET https://api.nuget.org/v3-flatcontainer/system.memory.data/index.json
info : GET https://api.nuget.org/v3-flatcontainer/system.text.encoding.web/index.json
info : OK https://api.nuget.org/v3-flatcontainer/system.valuetuple/4.5.0/system.valuetuple.4.5.0.nupkg 5ms
info : OK https://api.nuget.org/v3-flatcontainer/microsoft.bcl.hashcode/index.json 33ms
info : GET https://api.nuget.org/v3-flatcontainer/microsoft.bcl.hashcode/1.1.0/microsoft.bcl.hashcode.1.1.0.nupkg
info : OK https://api.nuget.org/v3-flatcontainer/microsoft.bcl.hashcode/1.1.0/microsoft.bcl.hashcode.1.1.0.nupkg 22ms
info : OK https://api.nuget.org/v3-flatcontainer/system.threading.tasks.extensions/index.json 380ms
```

```
portal.azure.com/#home
Microsoft Azure Upgrade
Search resources, services, and docs (G+/)
Copilot
mayureshkpatil50@gmail.com
DEFAULT DIRECTORY (MAYURESHK)

info : X.509 certificate chain validation will use the fallback certificate bundle at '/usr/share/dotnet/sdk/9.0.308/trustedroots/timestampctl.pem'.
info : Adding PackageReference for package 'Newtonsoft.Json' into project '/home/mayuresh/comosdb/comosdb.csproj'.
info : Restoring packages for /home/mayuresh/comosdb/comosdb.csproj...
info : CACHE https://api.nuget.org/v3-flatcontainer/newtonsoft.json/index.json
info : GET https://api.nuget.org/v3-flatcontainer/newtonsoft.json/index.json 378ms
info : OK https://api.nuget.org/v3-flatcontainer/newtonsoft.json/13.0.4/newtonsoft.json.13.0.4.nupkg
info : GET https://api.nuget.org/v3-flatcontainer/newtonsoft.json/13.0.4/newtonsoft.json.13.0.4.nupkg 13ms
info : Installed Newtonsoft.Json 13.0.4 from https://api.nuget.org/v3/index.json to /home/mayuresh/.nuget/packages/newtonsoft.json/13.0.4 with content hash pdgNNai3zv51W5ag268aujXUyx78Nde2bjlw2CwJaQrFMZFV2601lbgTopid2Gm67Ji1huLfwxc92qnfF/IC6A==.
info : CACHE https://api.nuget.org/v3/vulnerabilities/index.json
info : CACHE https://api.nuget.org/v3/vulnerabilities/2026.02.11.05.52.48/vulnerability.base.json
info : CACHE https://api.nuget.org/v3/vulnerabilities/2026.02.11.05.52.48/2026.02.11.11.52.48/vulnerability.update.json
info : Package 'Newtonsoft.Json' is compatible with all the specified frameworks in project '/home/mayuresh/comosdb/comosdb.csproj'.
info : PackageReference for package 'Newtonsoft.Json' version '13.*' added to file '/home/mayuresh/comosdb/comosdb.csproj'.
info : Writing assets file to disk. Path: /home/mayuresh/comosdb/obj/project.assets.json
log : Restored /home/mayuresh/comosdb/comosdb.csproj (in 930 ms).

Build succeeded in 0.7s
info : X.509 certificate chain validation will use the fallback certificate bundle at '/usr/share/dotnet/sdk/9.0.308/trustedroots/codesignctl.pem'.
info : X.509 certificate chain validation will use the fallback certificate bundle at '/usr/share/dotnet/sdk/9.0.308/trustedroots/timestampctl.pem'.
info : Adding PackageReference for package 'dotenv.net' into project '/home/mayuresh/comosdb/comosdb.csproj'.
info : GET https://api.nuget.org/v3/registration5-gz-semver2/dotenv.net/index.json 371ms
info : OK https://api.nuget.org/v3/registration5-gz-semver2/dotenv.net/index.json 274ms
info : CACHE https://api.nuget.org/v3-flatcontainer/microsoft.azure.common/index.json
info : CACHE https://api.nuget.org/v3-flatcontainer/newtonsoft.json/index.json
info : GET https://api.nuget.org/v3-flatcontainer/dotenv.net/index.json
info : OK https://api.nuget.org/v3-flatcontainer/dotenv.net/4.0.0/dotenv.net.4.0.0.nupkg 25ms
info : OK https://api.nuget.org/v3-flatcontainer/dotenv.net/4.0.0/dotenv.net.4.0.0.nupkg 25ms
info : Installed dotenv.net 4.0.0 from https://api.nuget.org/v3/index.json to /home/mayuresh/.nuget/packages/dotenv.net/4.0.0 with content hash 8M8Q6L/E3taLKDFv23M8W9yp511YqsXkylCjCjV022tUL336j1gV0Gms416YtcSB7j3Fw2dnddpEXnpNQ==.
info : CACHE https://api.nuget.org/v3/vulnerabilities/index.json
info : CACHE https://api.nuget.org/v3/vulnerabilities/2026.02.11.05.52.48/vulnerability.base.json
info : CACHE https://api.nuget.org/v3/vulnerabilities/2026.02.11.05.52.48/2026.02.11.11.52.48/vulnerability.update.json
info : Package 'dotenv.net' is compatible with all the specified frameworks in project '/home/mayuresh/comosdb/comosdb.csproj'.
info : PackageReference for package 'dotenv.net' version '4.0.0' added to file '/home/mayuresh/comosdb/comosdb.csproj'.
info : Writing assets file to disk. Path: /home/mayuresh/comosdb/obj/project.assets.json
log : Restored /home/mayuresh/comosdb/comosdb.csproj (in 669 ms).

mayuresh [ ~/comosdb ]$ touch .env
code .env
mayuresh [ ~/comosdb ]$ code Program.cs
mayuresh [ ~/comosdb ]$ dotnet build
Restore complete (0.5s)
comosdb succeeded (3.4s) -> bin/Debug/net9.0/comosdb.dll

Build succeeded in 4.2s
mayuresh [ ~/comosdb ]$
```

portal.azure.com/#home

Search resources, services, and docs (G+)

Copilot

mayureshkpatis0@gmail...
DEFAULT DIRECTORY (MAYURES...)

Welcome back, Mayuresh



Start with a template

Deploy in minutes using pre-made templates.



Create a resource

Choose a service to create a resource in your subscription.



Build an AI agent

Create and manage AI apps and agents using the latest models.



Import code from GitHub

Connect your GitHub account and deploy existing repositories.

₹18,351.57 in credit - Expires March 11, 2026

Bash

```
info : CACHE https://api.muget.org/v3-flatcontainer/newtonsoft.json/index.json
info : GET https://api.muget.org/v3-flatcontainer/dotenv.net/index.json
info : OK https://api.muget.org/v3-flatcontainer/dotenv.net/index.json 274ms
info : GET https://api.muget.org/v3-flatcontainer/dotenv.net/4.0.0/dotenv.net.4.0.0.mupkg
info : OK https://api.muget.org/v3-flatcontainer/dotenv.net/4.0.0/dotenv.net.4.0.0.mupkg 25ms
info : Installed dotenv.net 4.0.0 from https://api.muget.org/v3/index.json to /home/mayuresh/.muget/packages/dotenv.net/4.0.0 with content hash 3M0QpGL/E3taKKDFKv23M0Whyp511YqgXKylGDjCjV0222tULJ36j1gV0gMs4L6YtcSB7j2fw2dnDnpEXnpNG==.
info : CACHE https://api.muget.org/v3-vulnerabilities/index.json
info : CACHE https://api.muget.org/v3-vulnerabilities/2026.02.11.05.52.48/vulnerability.base.json
info : CACHE https://api.muget.org/v3-vulnerabilities/2026.02.11.05.52.48/2026.02.11.11.52.48/vulnerability.update.json
info : Package 'dotenv.net' is compatible with all the specified frameworks in project '/home/mayuresh/cosmosdb/cosmosdb.csproj'.
info : PackageReference for package 'dotenv.net' version '4.0.0' added to file '/home/mayuresh/cosmosdb/cosmosdb.csproj'.
info : Writing assets file to disk. Path: /home/mayuresh/cosmosdb/obj/project.assets.json
log : Restored /home/mayuresh/cosmosdb/cosmosdb.csproj (in 669 ms).
mayuresh [ ~/cosmosdb ]$ touch .env
code .env
mayuresh [ ~/cosmosdb ]$ code Program.cs
mayuresh [ ~/cosmosdb ]$ dotnet build
Restore complete (0.5s)
cosmosdb succeeded (3.4s) -> bin/Debug/net9.0/cosmosdb.dll

Build succeeded in 4.2s
mayuresh [ ~/cosmosdb ]$ code Program.cs
mayuresh [ ~/cosmosdb ]$ dotnet run
Created or retrieved database: mydatabase
Created or retrieved container: myContainer
Created item with ID: 360427b0-80d0-46c4-8131-dc4b70a90042
Request charge: 6.29 RU/s
mayuresh [ ~/cosmosdb ]$
```

Program.cs file

```
portal.azure.com/#home
Search resources, services, and docs (G+/)
Copilot
mayureshkpatil50@gmail...
DEFAULT DIRECTORY (MAYURES...)

Bash
Program.cs
1  using Microsoft.Azure.Cosmos;
2  using dotenv.net;
3
4  string databaseName = "myDatabase";
5  string containerName = "myContainer";
6
7  // Load environment variables from .env file
8  DotEnv.Load();
9  var envVars = DotEnv.Read();
10 string cosmosDbAccountUrl = envVars["DOCUMENT_ENDPOINT"];
11 string accountKey = envVars["ACCOUNT_KEY"];
12
13 if (string.IsNullOrEmpty(cosmosDbAccountUrl) || string.IsNullOrEmpty(accountKey))
14 {
15     Console.WriteLine("Please set the DOCUMENT_ENDPOINT and ACCOUNT_KEY environment variables.");
16     return;
17 }
18
19 // CREATE THE COSMOS DB CLIENT USING THE ACCOUNT URL AND KEY
20 CosmosClient client = new(
21     accountEndpoint: cosmosDbAccountUrl,
22     authKeyOrResourceToken: accountKey
23 );
24
25 try
26 {
27     // CREATE A DATABASE IF IT DOESN'T ALREADY EXIST
28     Database database = await client.CreateDatabaseIfNotExistsAsync(databaseName);
29     Console.WriteLine($"Created or retrieved database: {database.Id}");
30
31     // CREATE A CONTAINER WITH A SPECIFIED PARTITION KEY
32     Container container = await database.CreateContainerIfNotExistsAsync(
33         id: containerName,
34         partitionKeyPath: "/id"
35     );
36     Console.WriteLine($"Created or retrieved container: {container.Id}");
37
38     // DEFINE A TYPED ITEM (PRODUCT) TO ADD TO THE CONTAINER
39     Product newItem = new Product
40     {
41         id = Guid.NewGuid().ToString(),
42         name = "Sample Item",
43         description = "This is a sample item in my Azure Cosmos DB exercise."
44     };
45
46     // ADD THE ITEM TO THE CONTAINER
47     ItemResponse<Product> createResponse = await container.CreateItemAsync(
48         item: newItem,
49         partitionKey: new PartitionKey(newItem.id)
50     );
51
52     Console.WriteLine($"Created item with ID: {createResponse.Resource.id}");
53     Console.WriteLine($"Request charge: {createResponse.RequestCharge} RUs");
54 }
55 catch (CosmosException ex)
56 {
57     Console.WriteLine($"Cosmos DB Error: {ex.StatusCode} - {ex.Message}");
58 }
59 catch (Exception ex)
60 {
61     Console.WriteLine($"Error: {ex.Message}");
62 }
63
64 // Product class
65 public class Product
66 {
67     public string? id { get; set; }
68     public string? name { get; set; }
69     public string? description { get; set; }
70 }
71
```

Output :

The screenshot shows the Azure Cosmos DB Data Explorer interface. The left sidebar contains a navigation menu with options like Overview, Activity log, Access control (IAM), Tags, Diagnose and solve problems, Quick start, Data Explorer (selected), Mirroring in Fabric, Container Copy, Resource visualizer, Settings, Integrations, Containers, Monitoring, Automation, and Help. The main pane displays a tree view of the database structure: myDatabase > myContainer > Items. A single item is selected, showing its details in a table and a JSON representation. The JSON data is as follows:

```
{
  "id": "360427b0-80d0-46c4-8131-dc4b70a90042",
  "name": "Sample Item",
  "description": "This is a sample item in my Azure Cosmos DB e",
  "_rid": "JddBACIGzSBAAAAA==",
  "self": "db/JddBAA==/colls/JddBACIGzs=/docs/JddBACIGzsBAA",
  "_etag": "\/V61805702-0000-0700-0000-690c01000001",
  "attachments": "attachments/",
  "_ts": 1770829853
}
```

Exercise : Build and run a container image with Azure Container Registry Tasks

The screenshot shows the Azure portal home page with a 'Welcome back, Mayuresh' message. Below the message are four main action buttons: 'Start with a template', 'Create a resource', 'Build an AI agent', and 'Import code from GitHub'. At the bottom, there are sections for 'Alerts and service issues' and 'Costs'. Below the portal screenshot is a terminal window showing the following commands and output:

```
mayuresh [ ~ ]$ az group list --output table
Name                Location    Status
-----                -
rg-acr-lab          centralindia Succeeded

mayuresh [ ~ ]$ az acr list --output table
Name                Resource Group Location    SKU    LOGIN SERVER    CREATION DATE    ADMIN ENABLED
-----                -
mayureshacr2026    rg-acr-lab    centralindia Basic    mayureshacr2026.azurecr.io 2026-02-12T04:25:17Z True

mayuresh [ ~ ]$
```

```
Windows PowerShell
PS C:\Users\Mayuresh> docker tag mcr.microsoft.com/hello-world mayureshacr2026.azurecr.io/hello-world:v1
PS C:\Users\Mayuresh> docker push mayureshacr2026.azurecr.io/hello-world:v1
The push refers to repository [mayureshacr2026.azurecr.io/hello-world]
1b930d010525: Pushed
v1: digest: sha256:92c7f9c92844bbb5d0a101b22f7c2a7949e40f8ea90c8b3bc396879d95e899a size: 524
PS C:\Users\Mayuresh> az acr repository list --name mayureshacr2026 --output table
Result
-----
hello-world
PS C:\Users\Mayuresh> docker run mayureshacr2026.azurecr.io/hello-world:v1

Hello from Docker!
This message shows that your installation appears to be working correctly.

To generate this message, Docker took the following steps:
1. The Docker client contacted the Docker daemon.
2. The Docker daemon pulled the "hello-world" image from the Docker Hub.
   (amd64)
3. The Docker daemon created a new container from that image which runs the
   executable that produces the output you are currently reading.
4. The Docker daemon streamed that output to the Docker client, which sent it
   to your terminal.

To try something more ambitious, you can run an Ubuntu container with:
$ docker run -it ubuntu bash

Share images, automate workflows, and more with a free Docker ID:
https://hub.docker.com/

For more examples and ideas, visit:
```

portal.azure.com/#home

Microsoft Azure Upgrade Search resources, services, and docs (G+/) Copilot

mayureshkpatil50@gmail... DEFAULT DIRECTORY (MAYURES...

Welcome back, Mayuresh

₹18,351.57 in credit - Expires March 11, 2026

Start with a template
Deploy in minutes using pre-made templates.

Create a resource
Choose a service to create a resource in your subscription.

Build an AI agent
Create and manage AI apps and agents using the latest models.

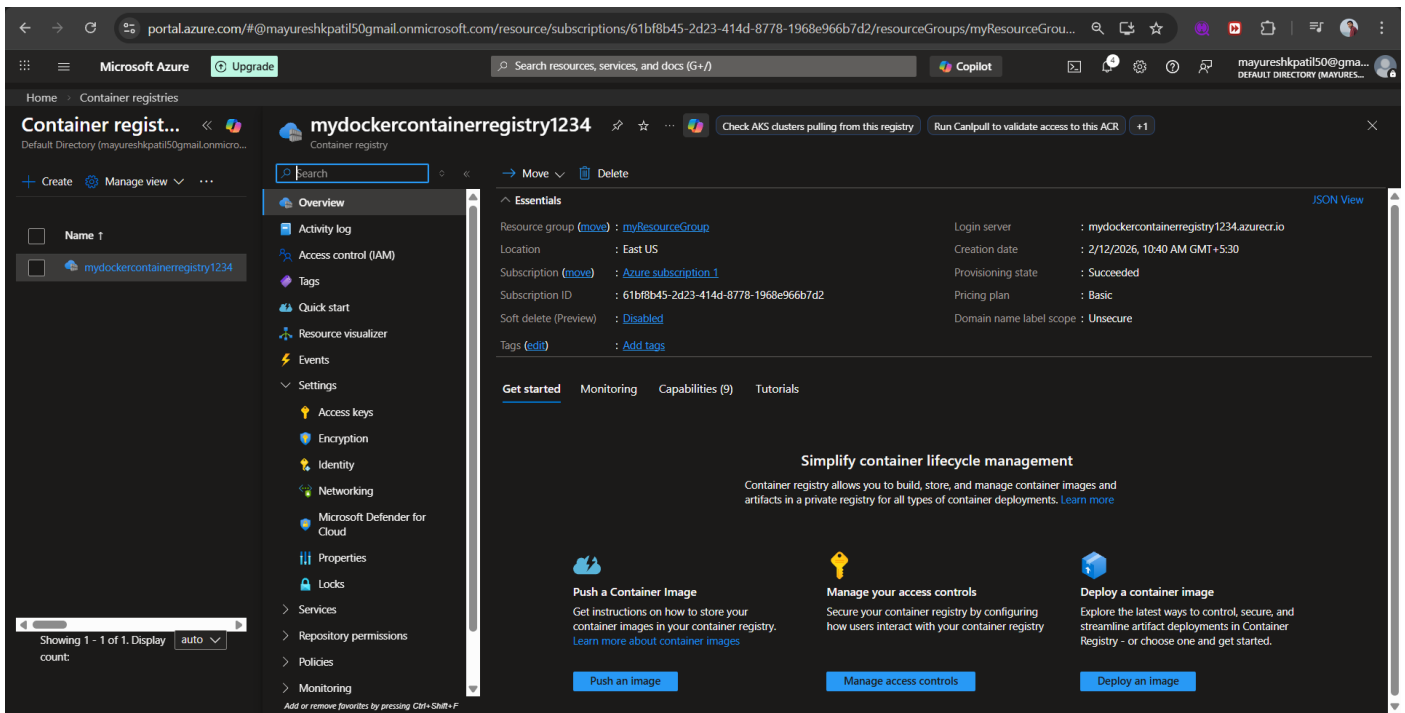
Import code from GitHub
Connect your GitHub account and deploy existing repositories.

Alerts and service issues
Last updated: a few seconds ago

Costs
Last updated: a few seconds ago

Switch to PowerShell Restart Manage files New session Editor Web preview Settings Help

mayuresh [~] \$ az acr repository list --name mydockercontainerregistry1234 --output table
Result
mayuresh [~] \$ az acr repository show-tags --name mydockercontainerregistry1234 --repository myapp --output table
Result
v1
mayuresh [~] \$ docker run mydockercontainerregistry1234.azurecr.io/myapp:v1
docker: Cannot connect to the Docker daemon at unix:///home/mayuresh/.docker/run/docker.sock. Is the docker daemon running?
See 'docker run --help'.
mayuresh [~] \$



Output : Successfully built and ran the Docker container locally, then pushed the image to Azure Container Registry.

```
C:\dockerlab>docker run mydockercontainerregistry1234.azurecr.io/myapp:v1
Hello from Docker!
This message shows that your installation appears to be working correctly.

To generate this message, Docker took the following steps:
 1. The Docker client contacted the Docker daemon.
 2. The Docker daemon pulled the "hello-world" image from the Docker Hub.
    (amd64)
 3. The Docker daemon created a new container from that image which runs the
    executable that produces the output you are currently reading.
 4. The Docker daemon streamed that output to the Docker client, which sent it
    to your terminal.

To try something more ambitious, you can run an Ubuntu container with:
$ docker run -it ubuntu bash

Share images, automate workflows, and more with a free Docker ID:
https://hub.docker.com/

For more examples and ideas, visit:
https://docs.docker.com/get-started/

C:\dockerlab>
```

Exercise - Deploy a container to Azure Container Instances using Azure CLI commands

```
portal.azure.com/#home
Microsoft Azure
Search resources, services, and docs (G+/)
Copilot
mayureshkpatil50@gmail...
DEFAULT DIRECTORY (MAYURES...)

Bash
Your Cloud Shell session will be ephemeral so no files or system changes will persist beyond your current session.
Requesting a Cloud Shell.Succeeded.
Connecting terminal...

mayuresh [ ~ ]$ az group create --location eastus --name myResourceGroup
{
  "id": "/subscriptions/61bf8b45-2d23-414d-8778-1968e9667d2/resourceGroups/myResourceGroup",
  "location": "eastus",
  "managedBy": null,
  "name": "myResourceGroup",
  "properties": {
    "provisioningState": "Succeeded"
  },
  "tags": null,
  "type": "Microsoft.Resources/resourceGroups"
}

mayuresh [ ~ ]$ az container create --resource-group myResourceGroup \
--name mycontainer \
--image mcr.microsoft.com/azuredocs/aci-helloworld \
--ports 80 \
--dns-name-label $INS_NAME_LABEL --location eastus \
--os-type Linux \
--cpu 1 \
--memory 1.5

{
  "confidentialComputeProperties": null,
  "containerGroupProfile": null,
  "containers": [
    {
      "command": null,
      "configMap": {
        "keyValuePairs": {}
      },
      "environmentVariables": {},
      "image": "mcr.microsoft.com/azuredocs/aci-helloworld",
      "instanceView": {
        "currentState": {
          "detailStatus": "",
          "exitCode": null,
          "finishTime": null,
          "startTime": "2026-02-12T05:54:02.677000+00:00",
          "state": "Running"
        },
        "events": [
          {
            "count": 1,
            "firstTimestamp": "2026-02-12T05:53:51+00:00",

```

```
portal.azure.com/#home
Microsoft Azure
Search resources, services, and docs (G+/)
Copilot
mayureshkpatil50@gmail...
DEFAULT DIRECTORY (MAYURES...)

Bash
"dnsConfig": null,
"encryptionProperties": null,
"extensions": null,
"id": "/subscriptions/61bf8b45-2d23-414d-8778-1968e9667d2/resourceGroups/myResourceGroup/providers/Microsoft.ContainerInstance/containerGroups/mycontainer",
"identity": null,
"imageRegistryCredentials": null,
"initContainers": [],
"instanceView": {
  "events": [],
  "state": "Running"
},
"ipAddress": {
  "autoGeneratedDomainNameLabelScope": "Unsecure",
  "dnsNameLabel": "aci-example-18016",
  "fqdn": "aci-example-18016.eastus.azurecontainer.io",
  "ip": "40.88.248.33",
  "ports": [
    {
      "port": 80,
      "protocol": "TCP"
    }
  ],
  "type": "Public"
},
"isCreatedFromStandbyPool": false,
"location": "eastus",
"name": "mycontainer",
"osType": "Linux",
"priority": null,
"provisioningState": "Succeeded",
"resourceGroup": "myResourceGroup",
"restartPolicy": null,
"sku": "Standard",
"standbyPoolProfile": null,
"subnetId": null,
"tags": {},
"type": "Microsoft.ContainerInstance/containerGroups",
"volumes": null,
"zones": null
}

mayuresh [ ~ ]$ az container show --resource-group myResourceGroup \
--name mycontainer \
--query "[?CIN:ipAddress.fqdn,ProvisioningState:provisioningState]" \
--out table

FQDN                                ProvisioningState
-----                                -
aci-example-18016.eastus.azurecontainer.io Succeeded

mayuresh [ ~ ]$
```



Welcome to Azure Container Instances!



Exercise : Deploy a container to Azure Container Apps with the Azure CLI

```
portal.azure.com/#home

Microsoft Azure Upgrade Search resources, services, and docs (G+/) Copilot mayureshkpatil50@gmail.com DEFAULT DIRECTORY (MAYURES...)

Switch to PowerShell Restart Manage files New session Editor Web preview Settings Help

Requesting a Cloud Shell. Succeeded.
Connecting terminal...

Your Cloud Shell session will be ephemeral so no files or system changes will persist beyond your current session.
mayuresh [ ~ ] $ az group create --location eastus --name myResourceGroup
{
  "id": "/subscriptions/61bf8b45-2d23-414d-8778-1968e96b7d2/resourceGroups/myResourceGroup",
  "location": "eastus",
  "managedBy": null,
  "name": "myResourceGroup",
  "properties": {
    "provisioningState": "Succeeded"
  },
  "tags": null,
  "type": "Microsoft.Resources/resourceGroups"
}
mayuresh [ ~ ] $ az extension add --name containerapp --upgrade
No stable version of 'containerapp' to install. Preview versions allowed.
The installed extension 'containerapp' is in preview.
mayuresh [ ~ ] $ az provider register --namespace Microsoft.App
Registering is still on-going. You can monitor using 'az provider show -n Microsoft.App'
mayuresh [ ~ ] $ az provider register --namespace Microsoft.App
mayuresh [ ~ ] $ az provider register --namespace Microsoft.OperationalInsights
mayuresh [ ~ ] $ az containerapp env create \
  --name my-container-env \
  --resource-group myResourceGroup \
  --location myLocation
The behavior of this command has been altered by the following extension: containerapp
You cannot create a Containerapp environment in location myLocation. List of eligible locations: westus2, southeastasia, swedencentral, canadacentral, westeurope, northeurope, eastus, eastus2, eastasia, australiaeast, germanywestcentral, japaneast, ukouth, westus, centralus, northcentralus, southcentralus, koreacentral, brasilsouth, westus3, francecentral, southafricanorth, norwayeast, switzerlandnorth, usenorth, canadacentral, westcentralus, ukwest, centralindia, japanwest, australiaeast, francecentral, spaincentral, italenorth, polandcentral, southindia.
mayuresh [ ~ ] $ az containerapp env create \
  --name my-container-env \
  --resource-group myResourceGroup \
  --location centralindia
The behavior of this command has been altered by the following extension: containerapp
No Log Analytics workspace provided.
Generating a Log Analytics workspace with name "workspace-mysourcecup@6f8"
Container Apps environment created. To deploy a container app, use: az containerapp create --help
{
  "id": "/subscriptions/61bf8b45-2d23-414d-8778-1968e96b7d2/resourceGroups/myResourceGroup/providers/Microsoft.App/managedEnvironments/my-container-env",
  "location": "Central India",
  "name": "my-container-env",
  "properties": {
    "appInsightsConfiguration": null,
    "appLogsConfiguration": {

```

```
portal.azure.com/#home

Microsoft Azure Upgrade Search resources, services, and docs (G+/) Copilot mayureshkpatil50@gmail.com DEFAULT DIRECTORY (MAYURES...)

Switch to PowerShell Restart Manage files New session Editor Web preview Settings Help

"eventStreamEndpoint": "https://centralindia.azurecontainerapps.dev/subscriptions/61bf8b45-2d23-414d-8778-1968e96b7d2/resourceGroups/myResourceGroup/managedEnvironments/my-container-env/eventstream",
  "infrastructureConfiguration": null,
  "ingressConfiguration": null,
  "kedaConfiguration": {
    "version": "2.17.2"
  },
  "openTelemetryConfiguration": null,
  "peerAuthentication": {
    "mtls": {
      "enabled": false
    }
  },
  "peerTrafficConfiguration": {
    "encryption": {
      "enabled": false
    }
  },
  "provisioningState": "Succeeded",
  "publicNetworkAccess": "Enabled",
  "staticIp": "20.207.94.116",
  "vnetConfiguration": null,
  "workloadProfiles": [
    {
      "enablePipe": false,
      "name": "Consumption",
      "workloadProfileType": "Consumption"
    }
  ],
  "zoneRedundant": false
},
"resourceGroup": "myResourceGroup",
"systemData": {
  "createdAt": "2026-02-12T06:12:25.363476",
  "createdBy": "mayureshkpatil50@gmail.com",
  "createdByType": "User",
  "lastModifiedAt": "2026-02-12T06:12:25.363476",
  "lastModifiedBy": "mayureshkpatil50@gmail.com",
  "lastModifiedByType": "User"
},
"type": "Microsoft.App/managedEnvironments"
}
mayuresh [ ~ ] $ az containerapp create \
  --name my-container-app \
  --resource-group myResourceGroup \
  --environment my-container-env \
  --image mcr.microsoft.com/azuredocs/containerapps-helloworld:latest \
  --target-port 80 \
  --ingress 'external' \
```

```
portal.azure.com/#home

Microsoft Azure Upgrade Search resources, services, and docs (G+/) Copilot mayureshkpatil50@gmail.com DEFAULT DIRECTORY (MAYURES...)

Switch to PowerShell Restart Manage files New session Editor Web preview Settings Help

{
  "mtls": {
    "enabled": false
  },
  "peerTrafficConfiguration": {
    "encryption": {
      "enabled": false
    }
  },
  "provisioningState": "Succeeded",
  "publicNetworkAccess": "Enabled",
  "staticIp": "20.207.94.116",
  "vnetConfiguration": null,
  "workloadProfiles": [
    {
      "enableFips": false,
      "name": "Consumption",
      "workloadProfileType": "Consumption"
    }
  ],
  "zoneRedundant": false
},
{
  "resourceGroup": "myResourceGroup",
  "systemData": {
    "createdAt": "2026-02-12T06:12:25.363476",
    "createdBy": "mayureshkpatil50@gmail.com",
    "createdByType": "User",
    "lastModifiedAt": "2026-02-12T06:12:25.363476",
    "lastModifiedBy": "mayureshkpatil50@gmail.com",
    "lastModifiedByType": "User"
  },
  "type": "Microsoft.App/managedEnvironments"
}
}

mayuresh [ ~ ]$ az containerapp create \
--name my-container-app \
--resource-group myResourceGroup \
--environment my-container-env \
--image mcr.microsoft.com/azuredocs/containerapps-helloworld:latest \
--target-port 80 \
--ingress 'external' \
--query properties.configuration.ingress.fqdn

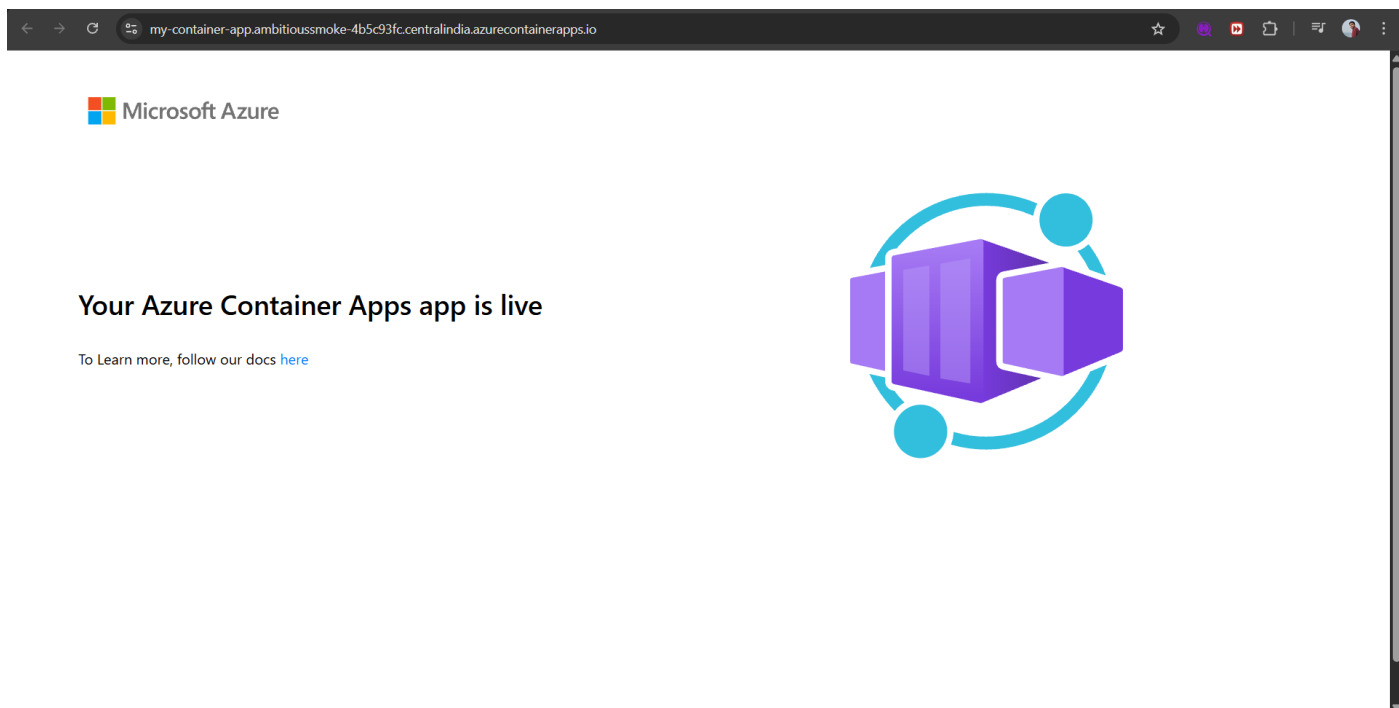
The behavior of this command has been altered by the following extension: containerapp

Container app created. Access your app at https://my-container-app.ambitiousmoke-4b5c93fc.centralindia.azurecontainerapps.io/

"my-container-app.ambitiousmoke-4b5c93fc.centralindia.azurecontainerapps.io"

mayuresh [ ~ ]$
```

Output :



Exercise - Implement interactive authentication with MSAL.NET

portal.azure.com/#view/Microsoft_AAD_RegisteredApps_MainBlade/~/Overview/appId/67024274-cc57-4464-85a7-2dac556e8b88/objectId/721e9c22-7...

Microsoft Azure Upgrade

Search resources, services, and docs (G+)

Copilot

mayureshkpatil50@gmail...
DEFAULT DIRECTORY (MAYURES...

Home > App registrations

myMsalApplication

Search

Delete Endpoints Preview features

Overview

Quickstart

Integration assistant

Diagnose and solve problems

Manage

Support + Troubleshooting

Essentials

Display name : myMsalApplication

Application (client) ID : 67024274-cc57-4464-85a7-2dac556e8b88

Object ID : 721e9c22-7583-4e13-9ae6-48edc7ed84c6

Directory (tenant) ID : b5f1c79d-ccb6-40ce-a57f-c6fe4b3e9b1d

Supported account types : My organization only

Client credentials : Add a certificate or secret

Redirect URIs : 0 web, 0 spa, 1 public client

Application ID URI : Add an Application ID URI

Managed application in L... : myMsalApplication

Welcome to the new and improved App registrations. Looking to learn how it's changed from App registrations (Legacy)? [Learn more](#)

Starting June 30th, 2020 we will no longer add any new features to Azure Active Directory Authentication Library (ADAL) and Azure Active Directory Graph. We will continue to provide technical support and security updates but we will no longer provide feature updates. Applications will need to be upgraded to Microsoft Authentication Library (MSAL) and Microsoft Graph. [Learn more](#)

Get Started Documentation

Build your application with the Microsoft identity platform

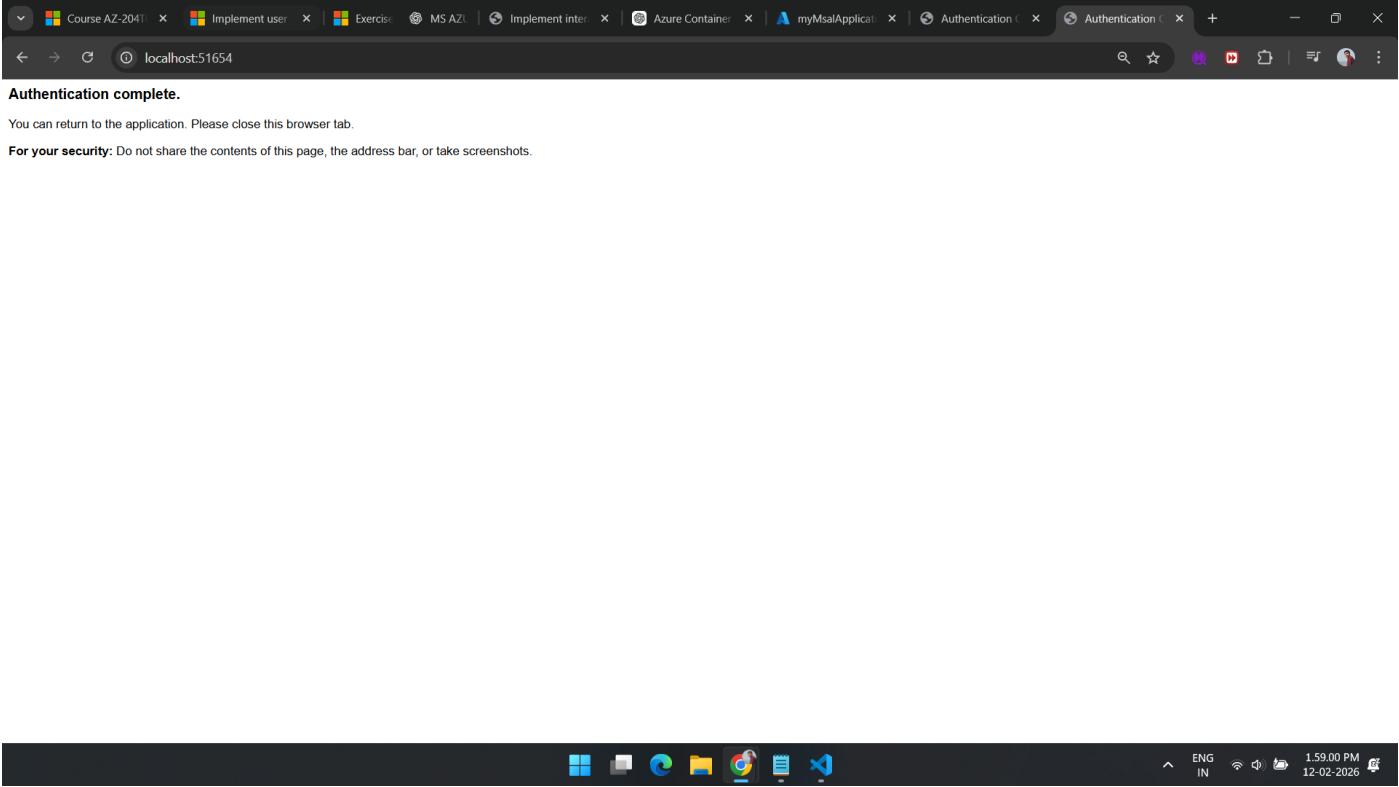
The Microsoft identity platform is an authentication service, open-source libraries, and application management tools. You can create modern, standards-based authentication solutions, access and protect APIs, and add sign-in for your users and customers. [Learn more](#)



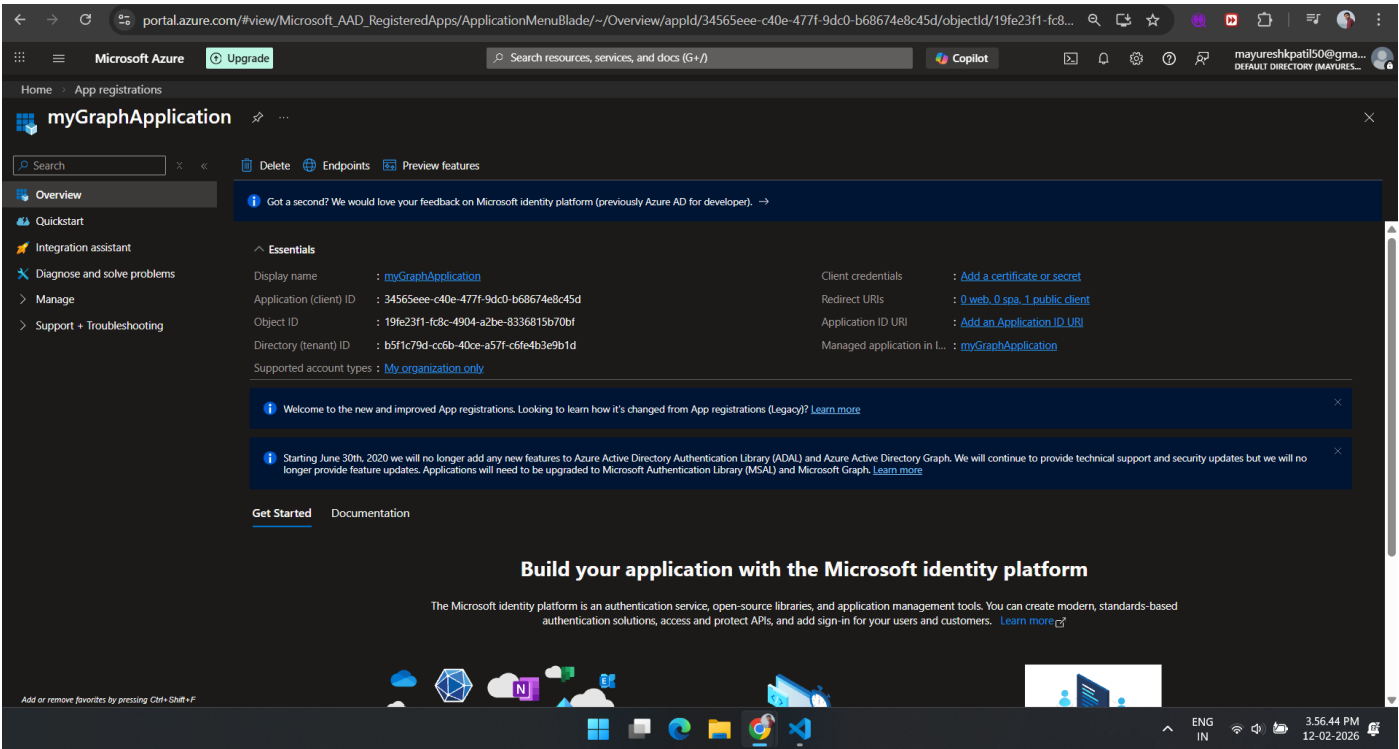




[illegible]



Exercise - Retrieve user profile information with the Microsoft Graph SDK



File Edit Selection View Go Run Terminal Help

THINKBRIDGE JANUARY 2026

Day 15

Day 16 Azure function

Day 17 Azure-3

Day 18

AuthApp

graphApp

bin

obj

env

graphApp.csproj

Program.cs

Day2

Day3

Day4

Day5

ThinkBridge_January_2026.sln

Day 18 > graphApp > Program.cs

```
1 using Microsoft.Graph;
2 using Azure.Identity;
3 using dotenv.net;
4
5 // Load environment variables from .env file (if present)
6 DotEnv.Load();
7 var envVars = DotEnv.Read();
8
9 // Read Azure AD app registration values
10 string clientId = envVars["CLIENT_ID"];
11 string tenantId = envVars["TENANT_ID"];
12
13 // Validate environment variables
14 if (string.IsNullOrEmpty(clientId) || string.IsNullOrEmpty(tenantId))
15 {
```

PROBLEMS OUTPUT DEBUG CONSOLE TERMINAL PORTS AZURE

PS C:\Users\Mayuresh\OneDrive\Desktop\ThinkBridge_January_2026\Day 18\graphApp> dotnet add package Azure.Identity

>> dotnet add package Microsoft.Graph

>> dotnet add package dotenv.net

info : GET https://api.nuget.org/v3/registration5-gz-semver2/dotenv.net/index.json

info : OK https://api.nuget.org/v3/registration5-gz-semver2/dotenv.net/index.json 420ms

info : Restoring packages for C:\Users\Mayuresh\OneDrive\Desktop\ThinkBridge_January_2026\Day 18\graphApp\graphApp.csproj...

info : CACHE https://api.nuget.org/v3/vulnerabilities/index.json

info : CACHE https://api.nuget.org/v3/vulnerabilities/2026.02.11.05.52.48/vulnerability.base.json

info : CACHE https://api.nuget.org/v3/vulnerabilities/2026.02.11.05.52.48/2026.02.12.05.18.10/vulnerability.update.json

info : Package 'dotenv.net' is compatible with all the specified frameworks in project 'C:\Users\Mayuresh\OneDrive\Desktop\ThinkBridge_January_2026\Day 18\graphApp\graphApp.csproj'.

info : PackageReference for package 'dotenv.net' version '4.0.0' added to file 'C:\Users\Mayuresh\OneDrive\Desktop\ThinkBridge_January_2026\Day 18\graphApp\graphApp.csproj'.

info : Writing assets file to disk. Path: C:\Users\Mayuresh\OneDrive\Desktop\ThinkBridge_January_2026\Day 18\graphApp\obj\project.assets.json

log : Restored C:\Users\Mayuresh\OneDrive\Desktop\ThinkBridge_January_2026\Day 18\graphApp\graphApp.csproj (in 155 ms).

PS C:\Users\Mayuresh\OneDrive\Desktop\ThinkBridge_January_2026\Day 18\graphApp> dotnet run

Retrieving user profile...

Display Name: Mayuresh Patil

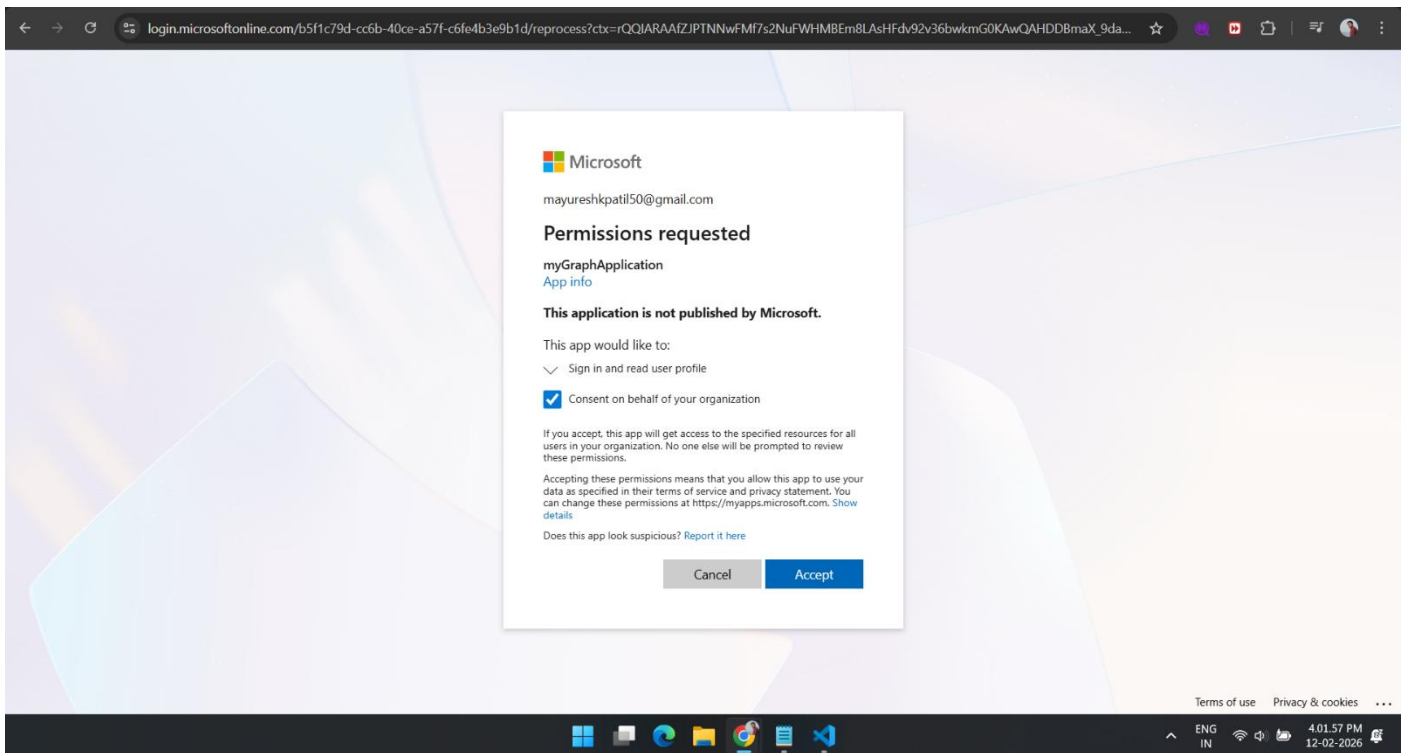
Principal Name: mayureshpatil50@gmail.com#EXT#@mayureshpatil50gmail.onmicrosoft.com

User Id: 2bffa31c1-0943-450c-ad41-7258d018ddf9

PS C:\Users\Mayuresh\OneDrive\Desktop\ThinkBridge_January_2026\Day 18\graphApp>

main* Launchpad 0 0 Projects: 1 Debug Any CPU Connect Solution file opened: ThinkBridge_January_2026.sln Ln 19, Col 1 Spaces: 4 UTF-8 with BOM CRLF C# Signed out Go Live Prettier

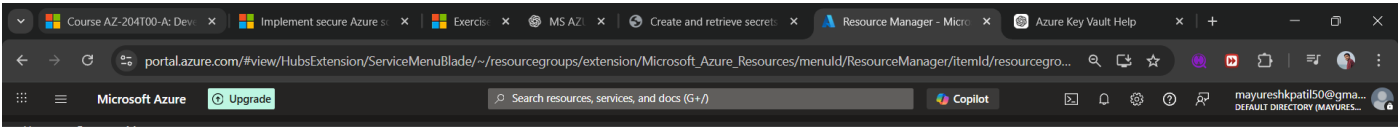
ENG IN 4:02:29 PM 12-02-2026



Authentication complete. You can return to the application. Feel free to close this browser tab.



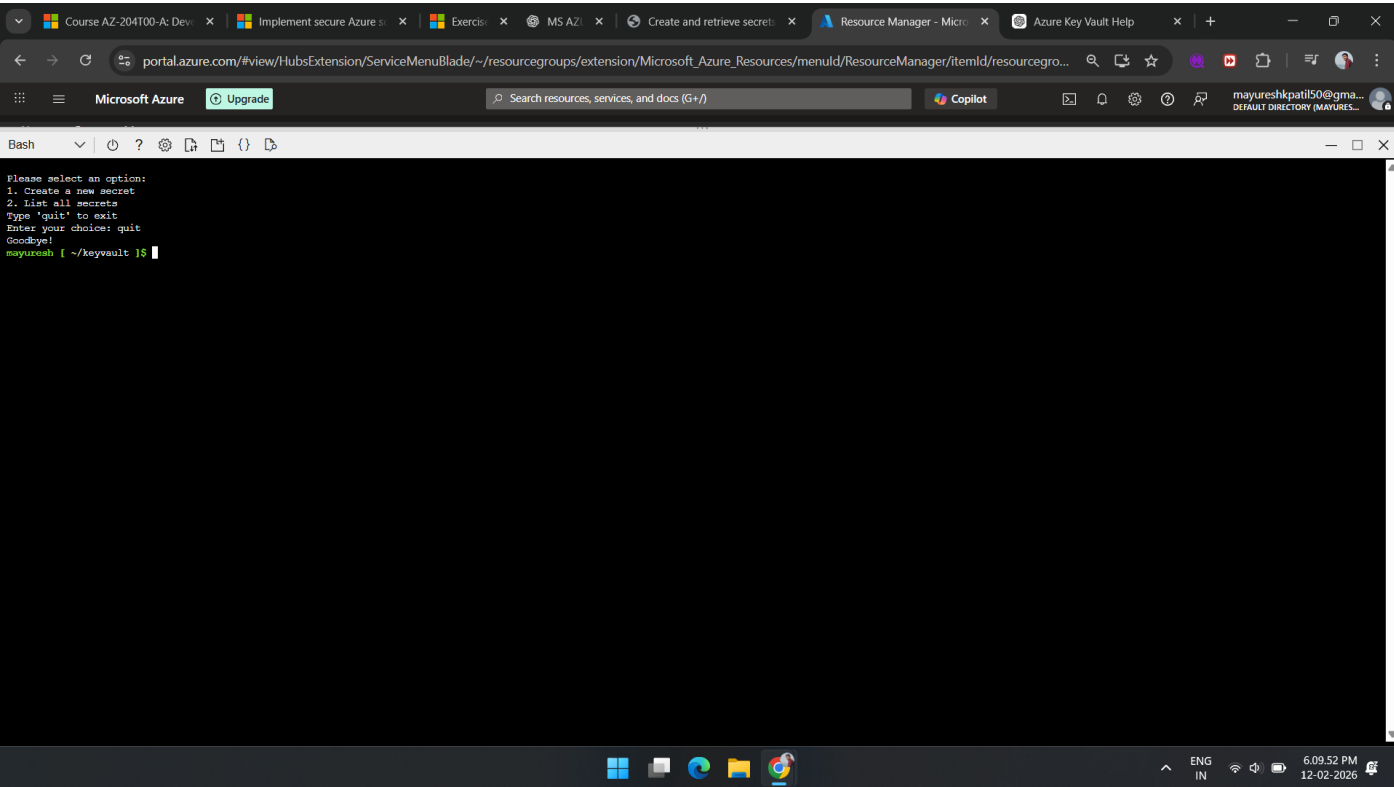
Exercise - Create and retrieve secrets from Azure Key Vault



```
Bash
mayuresh [ ~ ]$ az group create --name myResourceGroup --location eastus
{
  "id": "/subscriptions/61bf8b45-2d23-414d-8778-1968e96b7d2/resourceGroups/myResourceGroup",
  "location": "eastus",
  "managedBy": null,
  "name": "myResourceGroup",
  "properties": {
    "provisioningState": "Succeeded"
  },
  "tags": null,
  "type": "Microsoft.Resources/resourceGroups"
}
mayuresh [ ~ ]$ resourceGroup=myResourceGroup
location=eastus
keyVaultName=mykeyvaultnameSHANDOM
mayuresh [ ~ ]$ echo $keyVaultName
mykeyvaultname27708
mayuresh [ ~ ]$ az provider register --namespace Microsoft.KeyVault
mayuresh [ ~ ]$ az provider show --namespace Microsoft.KeyVault --query registrationState
"Registered"
mayuresh [ ~ ]$ az keyvault create \
--name $keyVaultName \
--resource-group $resourceGroup \
--location $location
{
  "id": "/subscriptions/61bf8b45-2d23-414d-8778-1968e96b7d2/resourceGroups/myResourceGroup/providers/Microsoft.KeyVault/vaults/mykeyvaultname27708",
  "location": "eastus",
  "name": "mykeyvaultname27708",
  "properties": {
    "accessPolicies": [],
    "createMode": null,
    "enableFurgeProtection": null,
    "enableBackupAuthorization": true,
    "enableSoftDelete": true,
    "enabledForDeployment": false,
    "enabledForDiskEncryption": false,
    "enabledForTemplateDeployment": false,
    "externalResourceId": null,
    "networkAcls": null,
    "privateEndpointConnections": null,
    "provisioningState": "Succeeded",
    "publicNetworkAccess": "Enabled",
    "sku": {
      "family": "A",
      "name": "standard"
    },
    "softDeleteRetentionInDays": 90,
  },
}
```

```
Course AZ-204100-A: Dev... Implement secure Azure s... Exercis... MS AZ... Create and retrieve secret... Resource Manager - Micro... Azure Key Vault Help...
portal.azure.com/#view/HubsExtension/ServiceMenuBlade/~/.resourcegroups/extension/Microsoft_Azure_Resources/menuld/ResourceManager/itemld/resourcegroups
Microsoft Azure Upgrade Search resources, services, and docs (G+/) Copilot mayureshkpatil50@gmail...
Bash
{"sku": {"family": "A", "name": "standard"}, "softDeleteRetentionInDays": 90, "tenantId": "155167d4-ecb3-40ce-a574-e6fe4b3e9b1d", "vaultUri": "https://mykeyvaultname27708.vault.azure.net/"}, "resourceGroup": "myResourceGroup", "systemData": {"createdAt": "2026-02-12T12:29:01.989000+00:00", "createdBy": "mayureshkpatil50@gmail.com", "createdByType": "User", "lastModifiedAt": "2026-02-12T12:29:01.989000+00:00", "lastModifiedBy": "mayureshkpatil50@gmail.com", "lastModifiedByType": "User"}, "tags": {}, "type": "Microsoft.KeyVault/vaults"}
mayuresh [ ~ ]$ userPrincipal=$(az rest --method GET --url https://graph.microsoft.com/v1.0/me \
--headers 'Content-Type:application/json' \
--query userPrincipalName --output tsv)
mayuresh [ ~ ]$ resourceID=$(az keyvault show --resource-group $resourceGroup \
--name $keyVaultName --query id --output tsv)
mayuresh [ ~ ]$ az role assignment create --assignee $userPrincipal \
--role "Key Vault Secrets Officer" \
--scope $resourceID
{
  "condition": null,
  "conditionVersion": null,
  "createdBy": null,
  "createdOn": "2026-02-12T12:30:43.587749+00:00",
  "delegatedManagedIdentityResourceId": null,
  "description": null,
  "id": "/subscriptions/61hfb845-2d23-414d-8778-1968e96b7d2/resourceGroups/myResourceGroup/providers/Microsoft.KeyVault/vaults/mykeyvaultname27708/providers/Microsoft.Authorization/roleAssignments/d13d00a0-8ada-412f-ab46-da5e6c79986",
  "name": "d13d00a0-8ada-412f-ab46-da5e6c79986",
  "principalId": "2bfff31c1-0943-450c-ad41-7258d018dd59",
  "principalType": "User",
  "resourceGroup": "myResourceGroup",
  "roleDefinitions": "/subscriptions/61hfb845-2d23-414d-8778-1968e96b7d2/providers/Microsoft.Authorization/roleDefinitions/b86a8f4-44ce-4948-aea5-ec6c2c155cd7",
  "scope": "/subscriptions/61hfb845-2d23-414d-8778-1968e96b7d2/resourceGroups/myResourceGroup/providers/Microsoft.KeyVault/vaults/mykeyvaultname27708",
  "type": "Microsoft.Authorization/roleAssignments",
  "updatedBy": "2bfff31c1-0943-450c-ad41-7258d018dd59",
  "updatedOn": "2026-02-12T12:30:45.545133+00:00"
}
https://portal.azure.com/#home
```

```
Course AZ-204100-A: Dev... Implement secure Azure s... Exercis... MS AZ... Create and retrieve secret... Resource Manager - Micro... Azure Key Vault Help...
portal.azure.com/#view/HubsExtension/ServiceMenuBlade/~/.resourcegroups/extension/Microsoft_Azure_Resources/menuld/ResourceManager/itemld/resourcegroups
Microsoft Azure Upgrade Search resources, services, and docs (G+/) Copilot mayureshkpatil50@gmail...
Bash
{"updatedBy": "2bfff31c1-0943-450c-ad41-7258d018dd59", "updatedOn": "2026-02-12T12:30:45.545133+00:00"}
mayuresh [ ~ ]$ az keyvault secret set --vault-name $keyVaultName \
--name "MySecret" --value "My secret value"
{
  "attributes": {
    "created": "2026-02-12T12:31:21+00:00",
    "enabled": true,
    "expires": null,
    "notBefore": null,
    "recoverableDays": 90,
    "recoveryLevel": "Recoverable+Purgeable",
    "updated": "2026-02-12T12:31:21+00:00"
  },
  "contentType": null,
  "id": "https://mykeyvaultname27708.vault.azure.net/secrets/MySecret/223c18c62e06401ca21549265e843d67",
  "kid": null,
  "managed": null,
  "name": "MySecret",
  "tags": {
    "file-encoding": "utf-8"
  },
  "value": "My secret value"
}
mayuresh [ ~ ]$ az keyvault secret show --name "MySecret" --vault-name $keyVaultName
{
  "attributes": {
    "created": "2026-02-12T12:31:21+00:00",
    "enabled": true,
    "expires": null,
    "notBefore": null,
    "recoverableDays": 90,
    "recoveryLevel": "Recoverable+Purgeable",
    "updated": "2026-02-12T12:31:21+00:00"
  },
  "contentType": null,
  "id": "https://mykeyvaultname27708.vault.azure.net/secrets/MySecret/223c18c62e06401ca21549265e843d67",
  "kid": null,
  "managed": null,
  "name": "MySecret",
  "tags": {
    "file-encoding": "utf-8"
  },
  "value": "My secret value"
}
mayuresh [ ~ ]$
```



Course AZ-204T00-A: Dev... Implement secure Azure s... Exercis... MS AZ... Create and retrieve secret... Resource Manager - Micro... Azure Key Vault Help

portal.azure.com/#view/HubsExtension/ServiceMenuBlade/~/_resourcegroups/extension/Microsoft_Azure_Resources/menuId/ResourceManager/itemId/resourcecegro...

Microsoft Azure Upgrade Search resources, services, and docs (G+/) Copilot mayureshkpatil50@gmail... DEFAULT DIRECTORY (MAYURES...

Bash Listing all secrets in the Key Vault...

Name: mySecret
Value: 147852369poiuytrmkjhgfdagwertyuio
Created: 2/12/2026 12:39:12 PM +00:00

Press Enter to continue...

6.09.35 PM 12-02-2026

Exercise : Retrieve configuration settings from Azure App Configuration

The screenshot shows the Microsoft Azure portal interface. At the top, there's a navigation bar with the 'Home - Microsoft Azure' tab selected. Below the navigation bar, a 'Welcome back, Mayuresh' message is displayed. To the right of the message, a credit balance of ₹18,351.57 is shown, expiring on March 11, 2026. Below the welcome message, there are four main action buttons: 'Start with a template', 'Create a resource', 'Build an AI agent', and 'Import code from GitHub'. Below these buttons, a terminal window is open, showing the following commands and output:

```
Bash
Requesting a Cloud Shell. Succeeded.
Connecting terminal...

Your Cloud Shell session will be ephemeral so no files or system changes will persist beyond your current session.
mayuresh [ ~ ] $ az group create --name myResourceGroup --location eastus
{
  "id": "/subscriptions/61bf8b45-2d23-414d-8778-1968e96b7d2/resourceGroups/myResourceGroup",
  "location": "eastus",
  "managedBy": null,
  "name": "myResourceGroup",
  "properties": {
    "provisioningState": "Succeeded"
  },
  "tags": null,
  "type": "Microsoft.Resources/resourceGroups"
}
mayuresh [ ~ ] $ az resourceGroup show --name myResourceGroup --location eastus
{
  "location": "eastus",
  "name": "myResourceGroup",
  "provisioningState": "Succeeded"
}
mayuresh [ ~ ] $ az appconfig create --name $appConfigName --resource-group $resourceGroup --sku Free
{
  "createMode": null,
  "creationDate": "2026-02-12T13:04:17+00:00",
  "dataPlaneProxy": {
    "authenticationMode": "Local",
    "privateLinkDelegation": "Disabled"
  },
  "defaultKeyValueRevisionRetentionPeriodInSeconds": 604800,
  "disableLocalAuth": false,
  "enablePurgeProtection": false,
  "encryption": {
    "keyVaultProperties": null
  },
  "endpoint": "https://appconfigname19290.azureappconfig.io",
  "id": "/subscriptions/61bf8b45-2d23-414d-8778-1968e96b7d2/resourceGroups/myResourceGroup/providers/Microsoft.AppConfiguration/configurationStores/appconfigname19290",
  "identity": null,
  "location": "eastus",
  "name": "appconfigname19290",
  "privateEndpointConnections": null,
  "provisioningState": "Succeeded",
  "publicNetworkAccess": null,
  "resourceGroup": "myResourceGroup",
  "sku": {
    "name": "Free"
  },
  "softDeleteRetentionInDays": 0,
  "systemData": {
    "createdAt": "2026-02-12T13:04:17+00:00",
    "createdBy": "mayureshkpatil50@gmail.com",
    "createdByType": "User",
    "lastModifiedAt": "2026-02-12T13:04:17+00:00",
    "lastModifiedBy": "mayureshkpatil50@gmail.com",
    "lastModifiedByType": "User"
  },
  "tags": {},
  "type": "Microsoft.AppConfiguration/configurationStores"
}
mayuresh [ ~ ] $ az provider show --namespace Microsoft.AppConfiguration --query "registrationState"
"Registering"
mayuresh [ ~ ] $ az provider show --namespace Microsoft.AppConfiguration --query "registrationState"
"Registering"
```

The screenshot shows the Microsoft Azure portal interface. At the top, there's a navigation bar with the 'Home - Microsoft Azure' tab selected. Below the navigation bar, a 'Welcome back, Mayuresh' message is displayed. To the right of the message, a credit balance of ₹18,351.57 is shown, expiring on March 11, 2026. Below the welcome message, there are four main action buttons: 'Start with a template', 'Create a resource', 'Build an AI agent', and 'Import code from GitHub'. Below these buttons, a terminal window is open, showing the following commands and output:

```
Bash
Requesting a Cloud Shell. Succeeded.
Connecting terminal...

Your Cloud Shell session will be ephemeral so no files or system changes will persist beyond your current session.
mayuresh [ ~ ] $ az provider show --namespace Microsoft.AppConfiguration --query "registrationState"
"Registered"
mayuresh [ ~ ] $ az appconfig create --location $location \
  --name $appConfigName \
  --resource-group $resourceGroup \
  --sku Free
{
  "createMode": null,
  "creationDate": "2026-02-12T13:04:17+00:00",
  "dataPlaneProxy": {
    "authenticationMode": "Local",
    "privateLinkDelegation": "Disabled"
  },
  "defaultKeyValueRevisionRetentionPeriodInSeconds": 604800,
  "disableLocalAuth": false,
  "enablePurgeProtection": false,
  "encryption": {
    "keyVaultProperties": null
  },
  "endpoint": "https://appconfigname19290.azureappconfig.io",
  "id": "/subscriptions/61bf8b45-2d23-414d-8778-1968e96b7d2/resourceGroups/myResourceGroup/providers/Microsoft.AppConfiguration/configurationStores/appconfigname19290",
  "identity": null,
  "location": "eastus",
  "name": "appconfigname19290",
  "privateEndpointConnections": null,
  "provisioningState": "Succeeded",
  "publicNetworkAccess": null,
  "resourceGroup": "myResourceGroup",
  "sku": {
    "name": "Free"
  },
  "softDeleteRetentionInDays": 0,
  "systemData": {
    "createdAt": "2026-02-12T13:04:17+00:00",
    "createdBy": "mayureshkpatil50@gmail.com",
    "createdByType": "User",
    "lastModifiedAt": "2026-02-12T13:04:17+00:00",
    "lastModifiedBy": "mayureshkpatil50@gmail.com",
    "lastModifiedByType": "User"
  },
  "tags": {},
  "type": "Microsoft.AppConfiguration/configurationStores"
}
mayuresh [ ~ ] $ az userPrincipal show --method GET --url https://graph.microsoft.com/v1.0/me \
  --headers 'Content-Type:application/json' \
  --query userPrincipalName --output tsv
mayuresh [ ~ ] $
```

```
Course AZ-204100-A: Develop... MS AZURE DAY 4... Retrieve configuration settings... Home - Microsoft Azure...
portal.azure.com/#home
Microsoft Azure Upgrade Search resources, services, and docs (G+/) Copilot
Bash
"lastModifiedByType": "User"
},
"tags": {},
"type": "Microsoft.AppConfiguration/configurationStores"
}
mayuresh [ ~ ] $ userPrincipal=$(az rest --method GET --url https://graph.microsoft.com/v1.0/me \
--headers 'Content-Type:application/json' \
--query userPrincipalName --output tsv)
mayuresh [ ~ ] $ resourceID=$(az appconfig show --resource-group $resourceGroup \
--name $appConfigName --query id --output tsv)
mayuresh [ ~ ] $ az role assignment create --assignee $userPrincipal \
--role "App Configuration Data Reader" \
--scope $resourceID
{
  "condition": null,
  "conditionVersion": null,
  "createdBy": null,
  "createdOn": "2026-02-12T13:05:37.134659+00:00",
  "delegatedManagedIdentityResourceId": null,
  "description": null,
  "id": "/subscriptions/61bf8b45-2d23-414d-8778-1968e96b7d2/resourceGroups/myresourcegroup/providers/Microsoft.AppConfiguration/configurationStores/appconfigname19290/providers/Microsoft.Authorization/roleAssignments/84b3221b-ab54-4cd4-aba2-1248252a4f69",
  "name": "84b3221b-ab54-4cd4-aba2-1248252a4f69",
  "principalId": "2bfff31c-0943-450c-ad41-7258d018ddF9",
  "principalType": "User",
  "resourceGroup": "myresourcegroup",
  "roleDefinitionId": "/subscriptions/61bf8b45-2d23-414d-8778-1968e96b7d2/providers/Microsoft.Authorization/roleDefinitions/516239f1-63e1-4d78-a4de-a74fb236a071",
  "scope": "/subscriptions/61bf8b45-2d23-414d-8778-1968e96b7d2/resourceGroups/myresourcegroup/providers/Microsoft.AppConfiguration/configurationStores/appconfigname19290",
  "type": "Microsoft.Authorization/roleAssignments",
  "updatedBy": "2bfff31c-0943-450c-ad41-7258d018ddF9",
  "updatedOn": "2026-02-12T13:05:38.523672+00:00"
}
mayuresh [ ~ ] $ az appconfig kv set --name $appConfigName \
--key Dev:connStr \
--value connectionstring \
--yes
{
  "contentType": "",
  "etag": "\"0C68F761D7F8FEE283C2W0gE-C2CF8ebabEQDjY\"",
  "key": "Dev:connStr",
  "label": null,
  "lastModified": "2026-02-12T13:05:58+00:00",
  "locked": false,
  "tags": {},
  "value": "connectionstring"
}
mayuresh [ ~ ] $
```

```
Course AZ-204100-A: Dev... MS AZURE DAY 4... Retrieve configuration sett... Home - Microsoft Azure... Inbox (4) - mayureshpati50@gmail... Azure App Configuration...
portal.azure.com/#home
Microsoft Azure Upgrade Search resources, services, and docs (G+/) Copilot
Bash
With the new Azure CLI login experience, you can select the subscription you want to use more easily. Learn more about it and its configuration at https://go.microsoft.com/fwlink/?linkid=2271236
If you encounter any problems, please open an issue at https://aka.ms/azclibug
[Warning] The login output has been updated. Please be aware that it no longer displays the full list of available subscriptions by default.
mayuresh [ ~ ] $ az login
/home/mayuresh/appconfig/Program.cs(18,20): warning CS8600: Converting null literal or possible null value to non-nullable type.
Error: ManagedIdentityCredential authentication failed: [Managed Identity] Error Code: AudienceNotSupported Error Message: Audience https://appconfig.azure.com is not a supported MSI token audience.
See the troubleshooting guide for more information. https://aka.ms/azadk/net/identity/managedidentitycredential/troubleshoot
mayuresh [ ~ ] $ az login
Cloud Shell is automatically authenticated under the initial account signed-in with. Run 'az login' only if you need to use a different account
To sign in, use a web browser to open the page https://microsoft.com/devicelogin and enter the code HNNB5GHS3 to authenticate.
^Cmayuresh [ ~ ] $ az login
mayuresh [ ~ ] $ az login
Cloud Shell is automatically authenticated under the initial account signed-in with. Run 'az login' only if you need to use a different account
To sign in, use a web browser to open the page https://microsoft.com/devicelogin and enter the code HNNB5GHS3 to authenticate.
Retrieving tenants and subscriptions for the selection...
[Tenant and subscription selection]
No Subscription name Subscription ID Tenant
[1] * Azure subscription 1 61bf8b45-2d23-414d-8778-1968e96b7d2 Default Directory
The default is marked with an *: the default tenant is 'Default Directory' and subscription is 'Azure subscription 1' (61bf8b45-2d23-414d-8778-1968e96b7d2).
Select a subscription and tenant (Type a number or Enter for no changes):
Tenant: Default Directory
Subscription: Azure subscription 1 (61bf8b45-2d23-414d-8778-1968e96b7d2)
[Announcements]
With the new Azure CLI login experience, you can select the subscription you want to use more easily. Learn more about it and its configuration at https://go.microsoft.com/fwlink/?linkid=2271236
If you encounter any problems, please open an issue at https://aka.ms/azclibug
[Warning] The login output has been updated. Please be aware that it no longer displays the full list of available subscriptions by default.
mayuresh [ ~ ] $ az login
Value from App Configuration:
connectionstring
mayuresh [ ~ ] $
```

Course AZ-204T00-A: Dev... MS AZURE DAY 4 Retrieve configuration sett... Mayuresh Patil - Microsoft... Inbox (4) - mayureshpatil... Azure App Configuration... learn.microsoft.com/en-us/training/courses/az-204t00

LEARNING PATH
Implement containerized solutions

3 Modules
Intermediate • Developer • Azure

COMPLETED ✓ Add

LEARNING PATH
Implement user authentication and authorization

4 Modules
Intermediate • Developer • Azure SDKs

Implement user authentication and... Add

COMPLETED ✓ Add

LEARNING PATH
Implement secure Azure solutions

3 Modules
Intermediate • Developer • Azure

COMPLETED ✓ Add

LEARNING PATH
Implement API Management

0 of 1 modules completed
Intermediate • Developer • Azure API Management

Start >

ENG IN 6:47:56 PM 12-02-2026

Course AZ-204T00-A: Dev... MS AZURE DAY 4 Retrieve configuration sett... Mayuresh Patil - Microsoft... Inbox (4) - mayureshpatil... Azure App Configuration... learn.microsoft.com/en-us/training/courses/az-204t00

Hide completed

LEARNING PATH
Implement Azure App Service web apps

4 Modules
Intermediate • Developer • Azure

COMPLETED ✓ Add

LEARNING PATH
Implement Azure Functions

2 Modules
Intermediate • Developer • Azure

COMPLETED ✓ Add

LEARNING PATH
Develop solutions that use Blob storage

3 Modules
Intermediate • Developer • Azure

COMPLETED ✓ Add

LEARNING PATH
Develop solutions that use Azure Cosmos DB

2 Modules
Intermediate • Developer • Azure

COMPLETED ✓ Add

ENG IN 6:47:41 PM 12-02-2026

Course AZ-204T00-A: Dev...MS AZURE DAY 4Retrieve configuration setti...Mayuresh Patil - MicrosoftInbox (4) - mayureshpatilAzure App Configuration...+portal.azure.com/#view/Microsoft_Azure_GTM/Billing.MenuView/~/overview/scopeld/%2Fproviders%2FMicrosoft.Billing%2FbillingAccounts%2F26279b87-121b-51b...Search resources, services, and docs (G+ /)Copilotmayureshpatil50@gma...DEFAULT DIRECTORY (MAYURES...

Home

Mayuresh Patil

Billing account

Search

Overview

Access control (IAM)

Billing scopes

Diagnose and solve problems

Cost management

Cost analysis

Cost alerts

Budgets

Advisor recommendations

Billing

Invoices

Payment methods

Payment history

Reservation transactions

Billing profiles

Benefits

Products + services

All billing subscriptions

Reservations + Hybrid

Add or remove favorites by pressing Ctrl+Shift+F

View Invoices

Spending rate and forecast

8

6

4

2

0

02/08

02/15

02/22

Current charges

₹6.81

Forecast

--

View details

This month's top products by charges

₹6.76

₹0.04

₹0.00

₹0.00

View details

Billing alerts (1)

Upgrade to keep going with your account

View alerts

Shortcuts

Download usage and prices

Manage payment methods

View billing subscriptions

Share monthly invoice by email

View your support plan

Give feedback

Credits remaining

₹18,351.57

0.03% of your total available credit used.

Used

Amount remaining

View credits

10.00

ENG

IN

6:47:25 PM

12-02-2026